

The Three-Center Electron-Repulsion Integral Is an Elliptic Bessel Moment: Genus One at the Third Center in Momentum-Space Slater Theory*

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(Dated: August 16, 2026)

Two-center electron-repulsion integrals over Slater-type orbitals close in elementary functions plus the exponential integral, the logarithm, and Euler’s γ —a genus-zero transcendence class—which is why the two-center problem has been solved since Roothaan and Ruedenberg. The genuine three-center integral has resisted a closed form for seventy years, and the standard verdict is that it is simply hard: reduce it to an infinite series or a one-dimensional quadrature and be done. We give it a structural reading instead. Evaluating the two-electron three-center integral $(XY|XZ)$ over $1s$ orbitals in momentum space—the Fock projection under which Coulomb–Sturmians become hyperspherical harmonics, the setting native to the Averys’ generalized-Sturmian program—the three centers enter as three *phases* rather than three foci, so the coordinate obstruction (prolate spheroidal coordinates admit exactly two foci) dissolves. **[MEASURED]** The resulting momentum integral reproduces the exact integral (agreement 1.9×10^{-14} against a closed-form Gaussian reference; 4.3×10^{-7} against an independent Slater evaluation), and its angular part collapses to a single spherical Bessel kernel. **[MEASURED / classical]** What remains—the radial content—is a two-scale Bessel moment whose period is a *complete elliptic integral* (verified to 31 digits), living on the genus-one curve $y^2 = (c_1 k^2 + 1)(c_2 k^2 + 1)$; it degenerates to the genus-zero two-center case exactly on the diagonal $c_1 = c_2$. **[SYMBOLIC + MEASURED]** The one-mass slice is the Laplace transform of the elliptic differential—an elliptic deformation of the Bessel K_0 , reducing to $K_0(D)$ when the two orbital scales coincide and to the complete elliptic integral K at zero separation—and satisfies an explicit fourth-order Picard–Fuchs equation with polynomial coefficients (residual $\sim 10^{-24}$), whose characteristic exponents are the four branch points of the curve. **[OBSERVATION]** This places a molecular integral inside the modern theory of Bessel moments and two-mass elliptic Feynman integrals; a targeted literature search finds that connection unmade in either field. **[OPEN]** We do *not* give the finite closed form, and we show the natural route to it is obstructed: the elliptic-dilogarithm (unequal-mass sunrise) mechanism does not engage, because the modulus Picard–Fuchs source is not an elementary subtopology but lies in the integral’s own Gauss–Manin D -module—the branch-point boundary that would supply the sunrise’s tadpole vanishes identically. **[OBSERVATION]** The object is therefore a genuinely new elliptic transcendent one level *above* the elliptic-dilogarithm tower, a period of a rank-four irregular connection; **[OPEN]** whether it admits any closed form is a differential-Galois question we leave open. The message is not a solution but a relocation: the third center does not make the radial integral messier within the two-center class, it raises the transcendence one genus, past the reach of the sunrise machinery.

I. THE TWO-CENTER CLASS, AND THE WALL BEYOND IT

The electron-repulsion integral over Slater-type orbitals on two centers is one of the oldest solved problems in molecular quantum mechanics. In prolate spheroidal coordinates, with the two nuclei at the foci, the Neumann expansion of $1/r_{12}$ terminates against the orbital angular momenta, and every integral reduces to elementary functions together with the exponential integral E_1 , the logarithm, and Euler’s constant γ . In the transcendence bookkeeping of the Geometric Vacuum program (Paper 18), these are *embedding-tier* constants—the cost of the continuum two-center geometry rather than content of the discrete graph—and, decisively, they sit at transcendence *weight one*: no π survives (it cancels against the Coulomb 4π), and no dilogarithm appears,

even though the ordered radial integral has the simplex shape that generically produces one. This weight-one closure is what lets the two-center repulsion density be *decided* rather than merely counted (Paper 58): the closed form is a rational combination of $\{1, e^{-\lambda R}\}$, and Lindemann–Weierstrass separates its zeros.

The genuine three-center integral has no such story. Prolate spheroidal coordinates are built from exactly two foci; a third nucleus has nowhere to sit. Gaussians escape this because two Gaussians on different centers multiply into a single Gaussian on a third point—Slater orbitals have no product theorem—and this, not any Geometric-Vacuum-specific limitation, is the seventy-year-old reason Gaussian bases displaced Slater orbitals for polyatomic quantum chemistry. The modern state of the art for three-center Slater repulsion integrals (Ref. [6]) reduces them to a single infinite expansion or a one-dimensional numerical quadrature; the one celebrated closed form in correlated two-electron theory, Fromm and Hill’s analytic evaluation [8], is a *dilogarithm*—a genus-zero object—and belongs to a different integral topology

* Paper 59 in the Geometric Vacuum series

(the one-center, r_{12} -correlated master integral), which lacks the structure we are about to exhibit.

This paper asks what the three-center integral *is*, transcendently, and answers: it is an elliptic object. The two-center weight-one closure is a genus-zero fact and it stops at two centers. The third center raises the transcendence one genus, to the elliptic (genus-one) world of Bessel moments and two-mass sunrise Feynman integrals. We establish this constructively, validate every step numerically, place the object in the literature, and mark the finite closed form as open.

Lineage and scope. The momentum-space route is not new machinery. Momentum-space Coulomb–Sturmians are hyperspherical harmonics under Fock’s conformal projection [1], the framework the Averys built into the generalized-Sturmian method [2, 3], and the closest prior art to the present calculation is the Averys’ own momentum-space evaluation of Coulomb–Sturmian repulsion integrals [4], which stops—as all prior work does—at expansion and numerics. What is new here is the reading: that the residual radial object is a specific elliptic Bessel moment, and hence that a molecular integral is an instance of the elliptic-Feynman-integral class. We work throughout with $1s$ orbitals at a common exponent; the generalization to higher (n, ℓ, m) multiplies the integrand by solid harmonics and shifts exponents but does not change the genus, and we return to this in Sec. VIII.

II. THE MOMENTUM-SPACE REDUCTION

Write the chemist integral for two densities $\rho_1 = \chi_X \chi_Y$ and $\rho_2 = \chi_X \chi_Z$, two two-center $1s$ densities that share the center X on different axes. With the Fourier representation of the kernel, $1/|\mathbf{r}_1 - \mathbf{r}_2| = (2\pi^2)^{-1} \int d^3k e^{i\mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2)} / k^2$, the integral becomes

$$(XY|XZ) = \frac{1}{2\pi^2} \int \frac{d^3k}{k^2} \tilde{\rho}_1(\mathbf{k}) \overline{\tilde{\rho}_2(\mathbf{k})}, \quad \tilde{\rho}(\mathbf{k}) = \int \rho(\mathbf{r}) e^{i\mathbf{k} \cdot \mathbf{r}} d^3r \quad (1)$$

Each center now appears only through a phase $e^{i\mathbf{k} \cdot \mathbf{R}}$: three centers are three phases, not three foci, and there is nothing to diagonalize. Equation (1) is exact; the only content is the density Fourier transform $\tilde{\rho}$ and the $\mathbf{k}$ integral.

We verify Eq. (1) two independent ways on the reference geometry $X = (0, 0, 0)$, $Y = (0, 0, 2)$, $Z = (1.5, 0, 0.5)$ (bohr), $\zeta = 1$, for which a McMurchie–Davidson evaluation over 8-Gaussian $1s$ fits gives the ground truth $(XY|XZ) = 0.204941722$. [**MEASURED**] Using the closed-form Fourier transform of the Gaussian-fitted density (a product of Gaussians is a Gaussian) in Eq. (1) reproduces the McMurchie–Davidson value on the *same* fits to 1.9×10^{-14} : the momentum formula, the Fourier and Coulomb conventions, and the $\mathbf{k}$ -quadrature are correct. Independently, computing the *true* Slater density transform (no Gaussian fit)

by a one-parameter reduction of the two-center Yukawa product reproduces the ground truth to 4.3×10^{-7} . Both are pinned in the regression suite [20].

The angular integral closes to a spherical Bessel kernel. Writing each Slater factor through $e^{-\zeta r} = -\partial_\zeta(e^{-\zeta r}/r)$ and the Feynman parametrization of the resulting Yukawa product, the three phases combine into a *single* phase $e^{i\mathbf{k} \cdot \mathbf{W}}$ with $\mathbf{W}(s, t) = (t-s)\mathbf{X} + s\mathbf{Y} - t\mathbf{Z}$ (s, t the two Feynman parameters), so the angular integral is elementary,

$$\int d\Omega_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{W}} = 4\pi j_0(k|\mathbf{W}|) = 4\pi \frac{\sin(k|\mathbf{W}|)}{k|\mathbf{W}|}. \quad (2)$$

The whole integral reduces to a two-dimensional Feynman integral over a one-dimensional radial (k) integral,

$$(XY|XZ) = \frac{8}{\pi} \partial_{\zeta_a} \partial_{\zeta_b} \partial_{\zeta_c} \partial_{\zeta_d} \int_0^1 ds \int_0^1 dt \int_0^\infty dk j_0(k|\mathbf{W}|) \times \frac{e^{-D_1 \Delta_1}}{\Delta_1} \frac{e^{-D_2 \Delta_2}}{\Delta_2} \Big|_{\zeta=1}, \quad (3)$$

with $\Delta_1 = \sqrt{c_1 k^2 + 1}$, $\Delta_2 = \sqrt{c_2 k^2 + 1}$, $c_1 = s(1-s)$, $c_2 = t(1-t)$, and $D_1 = |X - Y|$, $D_2 = |X - Z|$. The ∂_ζ ’s (a bookkeeping device restoring the Slater from the Yukawa) and the (s, t) integrations are elementary operations; the transcendental content is entirely in the radial factor. Equation (3) reproduces the ground truth to 1.8×10^{-6} [**MEASURED**], the residual being the finite-order angular and Feynman quadrature.

III. THE TRANSCENDENCE JUMP: GENUS ZERO TO GENUS ONE

The radial factor of Eq. (3) carries two dispersion factors Δ_1, Δ_2 with *different* scales $c_1 \neq c_2$. A single such factor is benign. Under Fock’s substitution $k\sqrt{c} = \frac{m}{\sinh \theta}$ it linearizes, and [**MEASURED**, 6×10^{-18}]

$$\int_0^\infty \cos(kb) \frac{e^{-D\sqrt{c k^2 + m^2}}}{\sqrt{c k^2 + m^2}} dk = \frac{1}{\sqrt{c}} K_0\left(\frac{m}{\sqrt{c}} \sqrt{c D^2 + b^2}\right), \quad (4)$$

a Bessel K_0 : the momentum-space Coulomb–Sturmian object, living on a rational (genus-zero) curve. This is why the two-center engine is weight one.

Two different scales cannot be linearized together. Their product is governed by the algebraic curve

$$y^2 = (c_1 k^2 + 1)(c_2 k^2 + 1), \quad (5)$$

a quartic with four distinct branch points $k = \pm i/\sqrt{c_1}, \pm i/\sqrt{c_2}$ whenever $c_1 \neq c_2$: an *elliptic curve*, genus one. When $c_1 = c_2$ the branch points collide and the quartic becomes a perfect square—rational, genus zero. The cleanest witness is the period. At $D_1 = D_2 = 0$ the radial integral is exactly a complete elliptic integral

[MEASURED, 31 digits; classical],

$$\int_0^\infty \frac{dk}{\sqrt{(c_1 k^2 + 1)(c_2 k^2 + 1)}} = \frac{1}{a_1 \sqrt{c_1 c_2}} K(m), \quad (6)$$

with $a_i = 1/\sqrt{c_i}$ and modulus $m = 1 - c_{\min}/c_{\max}$ (nondegenerate, $0 < m < 1$), while on the diagonal $c_1 = c_2 = c$ it collapses to the elementary $\pi/(2\sqrt{c})$. Since $c_1 = s(1-s)$ and $c_2 = t(1-t)$ range independently over $[0, \frac{1}{4}]$, the full integral (3) is an integral over a *family* of elliptic curves, of modulus $m(s, t) = 1 - c_{\min}/c_{\max}$, degenerate only on the measure-zero locus $s = t$ or $s = 1-t$. Every term generated by the ∂_ζ 's is a meromorphic function on the same curve (5), so the whole integral is a period of this elliptic family.

The structural statement is that the embedding tier of the transcendence taxonomy (Papers 18, 34) is *genus-graded*: genus zero ($\{E_1, \ln, \gamma\}$) at two centers, genus one (elliptic) at three. The two two-center negatives—no π , no dilogarithm—are genus-zero properties; the third center does not climb to a higher rung of the polylogarithm weight tower but leaves that tower for the elliptic one. This is the transcendence-level form of “no coordinate system for three foci,” and it also explains, after the fact, why an attempt to identify the three-center value against dilogarithm constants was structurally doomed: the constants are in the wrong class.

IV. THE REDUCTION: AN ELLIPTIC BESSEL MOMENT

The radial factor can be named precisely. Applying Eq. (4) to each factor and Parseval's theorem, the radial core (at fixed Feynman parameters, before the elementary j_0 and ∂_ζ wrapping) is a two-mass Bessel moment [MEASURED, 30 digits],

$$M = \frac{2}{\pi \sqrt{c_1 c_2}} \int_0^\infty db K_0(a_1 \sqrt{p_1^2 + b^2}) K_0(a_2 \sqrt{p_2^2 + b^2}), \quad (7)$$

with $a_i = 1/\sqrt{c_i}$ and $p_i = D_i \sqrt{c_i}$. This is a shifted-argument, two-scale member of the Bessel-moment family; it is not one of the single-scale moments $\int_0^\infty t^k K_0^n(t) dt$ catalogued in closed form by Bailey, Borwein, Broadhurst and Glasser [9].

A change of variable $x = \sqrt{c_1 k^2 + 1}$ on the one-mass slice $N(D) \equiv M(D, 0)$ puts it in the cleanest canonical form: the Laplace transform of the holomorphic elliptic differential,

$$N(D) = \frac{1}{\sqrt{c_1}} \int_1^\infty \frac{e^{-Dx} dx}{\sqrt{(x^2 - 1)(\rho x^2 + 1 - \rho)}}, \quad \rho = \frac{c_2}{c_1}. \quad (8)$$

This is a single explicit one-dimensional integral of an elementary integrand—already sharper than the infinite series of the standard treatment [6], and it makes the elliptic structure manifest. It is an *elliptic deformation* of

the Bessel K_0 : as $\rho \rightarrow 0$ (coincident scales) $N \rightarrow K_0(D)$, and at $D = 0$ it is the complete elliptic integral K of Eq. (6).

Because $1/\sqrt{Q}$, $Q(x) = (x^2 - 1)(\rho x^2 + 1 - \rho)$, satisfies the first-order relation $Qf' + \frac{1}{2}Q'f = 0$, and the boundary at the branch point $x = 1$ vanishes ($Q(1)^{1/2} = 0$), integration by parts against e^{-Dx} (using $\int_1^\infty x^n e^{-Dx} f dx = (-1)^n L^{(n)}(D)$, $L \equiv \sqrt{c_1} N$) yields an exact fourth-order Picard–Fuchs equation with polynomial coefficients [SYMBOLIC PROOF; MEASURED residual $\sim 10^{-24}$],

$$D\rho L'''' + 2\rho L''' + D(1-2\rho)L'' + (1-2\rho)L' - D(1-\rho)L = 0. \quad (9)$$

Its characteristic polynomial factors as $\rho(s^2 - 1)(s^2 + (1 - \rho)/\rho)$, whose roots $s = \pm 1$ and $s = \pm i\sqrt{(1 - \rho)/\rho}$ are exactly the four branch points of the curve (5): the K_0/I_0 sector from the $(x^2 - 1)$ pair, the oscillatory J_0/Y_0 sector from the other. A subordination identity casts the same object as a two-scale sunrise integral, $\propto \iint ds_1 ds_2 (s_1 s_2)^{-1/2} e^{-D_1^2/4s_1 - s_1 - D_2^2/4s_2 - s_2} / \sqrt{s_1 c_1 + s_2 c_2}$, whose second-Symanzik-type $\sqrt{\text{linear}}$ form is the genus-one signature and separates only when $c_1 = c_2$.

One numerical fact pins the object's class beyond the period. $N(D)$ is *non-analytic* in D at $D = 0$: it carries a $D \ln D$ term [MEASURED], so it is not a pure elliptic integral (K/E are analytic in their parameters). The elliptic $D = 0$ period together with the $D \ln D$ non-analyticity are the two fingerprints of an *elliptic* $\oplus$ *logarithmic* object—the inhomogeneous (“regulator”) term of the Picard–Fuchs solution associated with the elliptic dilogarithm [11, 12]—a correspondence whose reach we test, and bound, in Sec. V.

V. THE ELLIPTIC-DILOGARITHM ROUTE IS OBSTRUCTED

The unequal-mass sunrise is closed by variation of parameters: a second-order Picard–Fuchs operator in the external variable annihilates the two periods, its inhomogeneity is an elementary subtopology (a tadpole), and the particular solution—an iterated integral of the periods against that elementary source—is the elliptic dilogarithm [11, 12]. Running the same machinery on $N(D)$ is the natural attempt. It does not close, and the reason is structural.

The curve $y^2 = Q(x, \rho)$ of Eq. (5) moves with the modulus ρ , not with the Laplace variable D . Accordingly the two periods $K(\rho)$, $K(1-\rho)$ are the homogeneous solutions of the second-order modulus operator [MEASURED: both annihilated to 10^{-18} ; $L(0, \rho) = K(1 - \rho)$ to 19 digits]

$$\mathcal{M}_\rho = \rho(1 - \rho) \partial_\rho^2 + (1 - 2\rho) \partial_\rho - \frac{1}{4}, \quad (10)$$

and they are *not* solutions of the fourth-order operator in D : a constant gives $\mathcal{O}[1] = -D(1 - \rho) \neq 0$, and the

single Bessel functions of both sectors, $K_0(D)$, $I_0(D)$ and $J_0(\omega D)$, $Y_0(\omega D)$ with $\omega = \sqrt{(1-\rho)/\rho}$, are likewise not D -solutions [**MEASURED**] (they are coupled, which is why Eq. (9) does not factor into elementary Bessel operators). The elliptic-dilogarithm construction lives on the periods, so it must be run in the modulus ρ .

There it meets the obstruction. Acting with $\mathcal{M}_\rho$ under the integral sign is an exact total derivative [**SYMBOLIC PROOF**; **exact residual** 0],

$$\mathcal{M}_\rho[Q^{-1/2}] = \frac{d}{dx} \left[g(x) Q^{-1/2} \right], \quad g(x) = \frac{-x(x^2 - 1)}{4(\rho x^2 + 1 - \rho)}, \quad (11)$$

so for $D \neq 0$ the modulus source is $S(D, \rho) \equiv \mathcal{M}_\rho[L] = D \int_1^\infty e^{-Dx} g Q^{-1/2} dx$. Because g vanishes at the branch point $x = 1$, differentiating in the modulus produces no branch-point boundary term: the analogue of the sunrise’s pinched propagator—the elementary tadpole that would be the inhomogeneity—is absent, and the source stays on the elliptic curve. Quantitatively [**MEASURED**, 45 **digits**], S closes exactly and finitely only inside the integral’s own Gauss–Manin D -module,

$$S(D, \rho) = \sum_{k=0}^3 P_k(D, \rho) \partial_D^k L, \quad \deg_D P_k = 2, \quad (12)$$

with held-out residual 5×10^{-44} , flat under increase of the fit degree (numerically an exact finite identity); whereas the genuine variation-of-parameters shape—period level $\{L, \partial_D L\}$ plus an elementary Bessel subtopology $\{K_0, K_1\}(D)$, $\{J_0, Y_0\}(\omega D)$ —only *approximates* it (residual 2×10^{-14} falling geometrically to 3×10^{-30} across fit degrees, never saturating). The inhomogeneity is in-module, not a simpler subtopology, and variation of parameters against the periods is therefore circular.

[**OBSERVATION**] $N(D)$ is thus one transcendence level above the sunrise. It is not an elliptic dilogarithm of elementary or period-level arguments; it is a period of the rank-four irregular connection defined by Eq. (9), for which the sunrise mechanism supplies no reduction to the elliptic-polylogarithm tower. Whether it admits a closed form of any kind is a differential-Galois question—whether the fourth-order operator factors over the differential field generated by the periods $K(\rho)$, $K(1-\rho)$ —which we leave open.

VI. PLACEMENT IN THE LITERATURE

The object of Eqs. (7)–(9) sits inside a fully developed body of mathematics, applied so far only to problems far from chemistry. Two-mass elliptic Feynman integrals—the sunrise/banana family—are evaluated in terms of elliptic dilogarithms as inhomogeneous solutions of exactly this kind of Picard–Fuchs equation [11–13]; their periods are the complete elliptic integrals of Eq. (6), and

at special (complex-multiplication) scale ratios their values reach critical L -values of modular forms [14, 15]. The closest published analog to Eq. (7) is a doubled-argument Bessel moment that Broadhurst evaluated as a product of complete elliptic integrals via a lattice-Green-function contour [10]: the same “two-scale moment $\rightarrow$ product of K ’s” mechanism we expect at rational ρ .

On the chemistry side the treatment is uniformly numerical: three-center Slater repulsion integrals are reduced to infinite series or one-dimensional quadrature [6, 7], and the Averys’ momentum-space Coulomb–Sturmian route [4], closest in spirit to ours, likewise stops at expansion and numerics. We could find no paper drawing the bridge in either direction: no chemistry reference names an electron-repulsion integral as a Bessel moment or an elliptic Feynman integral, and no amplitudes reference applies the elliptic machinery to a molecular integral. [**OBSERVATION**] The identification—that the three-center Slater integral, via the Fock projection, is a two-scale Bessel moment on the sunrise elliptic curve—appears to be unmade. This is an absence from a targeted search, not a proof of absence; the residual risk is an obscure single source.

VII. THE MODULAR STRUCTURE AND ITS COSMIC-GALOIS PERIODS

The genus-one curve of Eq. (5) is not an arbitrary elliptic curve: the family over the scale-ratio base is a *modular* family, and its periods are periods of a specific modular curve. Forming the modular parameter from the two periods of Sec. V, $\tau(\rho) = iK(\rho)/K(1-\rho)$, the theta-series modular lambda satisfies [**MEASURED**, 10^{-41} ; **classical**]

$$\lambda(\tau(\rho)) = 1 - \rho, \quad (13)$$

an exact, rational—indeed linear—modulus map. The family is therefore the Legendre family, the universal elliptic curve over the modular curve $X(2) = \Gamma(2) \backslash \mathfrak{H}^*$, pulled back along $\lambda = 1 - \rho$; it is non-isotrivial ($j(\rho)$ varies) with three cusps at $\rho \in \{0, 1, \infty\}$ matching the three cusps of $\Gamma(2)$ (the cusp $\rho = 1$ is the coincident-scale diagonal). The periods $K(1-\rho)$, $K(\rho)$ are the $\Gamma(2)$ modular periods.

Since $\rho = c_2/c_1$ ranges over all of $(0, \infty)$ as the Feynman parameters range over $(0, 1)$, the parameter integral sweeps the entire real locus of $X(2)$ and passes through every complex-multiplication fibre. There the periods are Γ -values by Chowla–Selberg [**MEASURED**, $\sim 10^{-51}$; **classical**]: at $\tau = i$ (the fibre $\rho = \frac{1}{2}$, *inside* the physical domain), $K(\frac{1}{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$; at $\tau = i\sqrt{2}$ (discriminant -8), $K = (1 + \sqrt{2})^{1/2} \Gamma(1/8) \Gamma(3/8)/(2^{13/4} \sqrt{\pi})$. A molecular three-center integral thus has, at the fibres its own parameter integral visits, periods that are classical Γ -values—across distinct fundamental discriminants, so the arithmetic tracks the fibre rather than a single coincidence.

The integrated integral is an elliptic multiple modular value. The full parameter integral pulls back, via Eq. (13), to an iterated integral over $X(2)$ of period-weighted forms—a multiple modular value for $\Gamma(2)$ in the sense of Brown, i.e. an elliptic polylogarithm. A dedicated evaluator (a fixed Gauss–Legendre tensor grid with a $\sin^2$ substitution that removes the endpoint non-analyticity, replacing the divergent-cost adaptive quadrature) gives the collinear value [MEASURED, ~ 17 digits] $0.39535576590171392\dots$; a weight-graded search finds it is *not* a low-height combination of the central fibre’s period $K(\frac{1}{2})$, its quasi-period $E(\frac{1}{2})$, and π , nor of the classical polylogarithms, consistent with a genuine (non-classical, multi-fibre) elliptic multiple modular value. [OBSERVATION] This places the three-center integral inside the elliptic (genus-one) layer of the cosmic-Galois / mixed-elliptic-motive structure that the Geometric Vacuum program’s Tannakian substrate reaches for [18]—the concrete elliptic period the mixed-Tate substrate could not itself supply. [OPEN] Its explicit realization in the $\Gamma(2)$ iterated-Eisenstein basis is the natural next step, of a piece with the finite closed form of Sec. V. We note the level is $\Gamma(2)$ —the classical *universal* Legendre level, below the two-loop sunrise’s $\Gamma_1(6)$; landing on the Legendre family is expected for a four-branch-point curve, so the content here is the rational modulus map and the in-domain complex-multiplication fibres, not an exotic level.

VIII. SCOPE, AND THE OPEN FRONTIER

Scope. The demonstration is for $1s$ orbitals at a common exponent. Higher (n, ℓ, m) multiply the integrand of Eq. (3) by solid-harmonic polynomials in $\mathbf{k}$ and shift the exponents; these change the weights and arguments but not the genus of the curve (5), which is set by the two distinct scales. The result is a statement about the transcendence *class* of the three-center integral, established on its simplest genuine instance. It does not by itself deliver a polyatomic energy: a molecular Hamiltonian requires the full three-center tensor, and a value that is part closed-form and part numerical is neither more accurate (the Gaussian fits are already exact to fit

quality) nor decidable. What the reduction buys is an exact, fast, geometry-native evaluator for the three-center integral and—the point of this paper—a precise diagnosis of why it is hard.

The open problem. As shown in Sec. V, the elliptic-dilogarithm route is obstructed: $N(D)$ is not an elliptic dilogarithm of elementary or period-level data but a period of the rank-four irregular connection (9). The remaining question is whether that connection admits *any* closed form—equivalently, whether its fourth-order operator factors over the differential field generated by the periods $K(\rho), K(1 - \rho)$ —a differential-Galois problem we do not settle. (This tempers a literature-scout expectation that the object would sit at the tractable elliptic-dilogarithm level because it has only two propagator factors; that pattern-match from the sunrise is overridden by the in-module computation of Sec. V.) Because it continues the Averys’ momentum-space Sturmian program and requires elliptic-Feynman-integral technique one level beyond the sunrise, it is the natural place for collaboration.

IX. CONCLUSION

The three-center electron-repulsion integral is not merely a harder version of the two-center one; it is an object of higher genus. Two centers close at genus zero over $\{E_1, \ln, \gamma\}$; three centers, read in momentum space where the coordinate obstruction dissolves, reduce to a two-scale Bessel moment on an elliptic curve, an elliptic deformation of the Bessel K_0 with an explicit fourth-order Picard–Fuchs equation and a period that is a complete elliptic integral. The seventy-year difficulty of polyatomic Slater integrals is, at the level of transcendence, a genus jump; and a classical molecular integral turns out to be an instance of the modern theory of elliptic Feynman integrals—a bridge that, to our knowledge, had not been drawn. The finite closed form remains open—and, we have shown, past the reach of the sunrise’s elliptic-dilogarithm mechanism, whose inhomogeneity is absent here: the object is a period of a rank-four irregular connection, and whether it closes at all is a differential-Galois question, posed here as the frontier it deserves to be.

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